

1	(a) work done bringing unit mass from infinity (to the point)	M1 A1	[2]
	(b) $E_P = -m\phi$	B1	[1]
	(c) $\phi \propto 1/x$	C1	
	either at $6R$ from centre, potential is $(6.3 \times 10^7)/6$ ($= 1.05 \times 10^7 \text{ J kg}^{-1}$)		
	and at $5R$ from centre, potential is $(6.3 \times 10^7)/5$ ($= 1.26 \times 10^7 \text{ J kg}^{-1}$)	C1	
	change in energy = $(1.26 - 1.05) \times 10^7 \times 1.3$	C1	
	$= 2.7 \times 10^6 \text{ J}$	A1	
	or change in potential = $(1/5 - 1/6) \times (6.3 \times 10^7)$	(C1)	
	change in energy = $(1/5 - 1/6) \times (6.3 \times 10^7) \times 1.3$	(C1)	
	$= 2.7 \times 10^6 \text{ J}$	(A1)	[4]